

1. Executive Summary

		Traffic Summary		Count	
		Total Flows Observed		76976	
		Total Packets Analyzed		5528219269	
		Malicious Packets		5507160908	
		True Negatives Passed		52123	
		Attack Vector		Count	
		ICMP Flood		6	
		SYN Flood		17	
		UDP Flood		7	
		Uncertain		14	
		Action		Count	
		Quarantined		10	
		Time Ban		10	
		Blackhole		0	
		Released		7	

Metric	Value
Total Threats Mitigated	44
High Priority	9
Low Priority	35
Manual Releases	0
Manual Blocks	0
ML Metrics	Value
False Positives	74
FP Rate	0.15%
IF Precision	99.69%
IF Recall	92.70%
IF F1-Score	96.07%
IF Accuracy	97.46%

2. Performance Benchmark

2a. Isolation Forest-Anomaly Detection

Metric	Value	Description
Precision	99.69%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	92.70%	Share of actual attacks successfully flagged
F1-Score	96.07%	Balanced measure combining Precision and Recall
Accuracy	97.46%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.15%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	7.30%	Actual attacks that went undetected
True Positive Rate (TPR)	92.70%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.85%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 23514	FN 1852
Actual: Normal	FP 74	TN 50271

2b. Random Forest-Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	13480	0	0
Act: ICMP	0	3580	0
Act: UDP	0	0	6454

2c. Response Latency

Metric	Value	Description
Detection Time	679.9 ms	Average time taken to detect an attack
Mitigation Response Time	15.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.87%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	26.85%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	9.31%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

No offence history for this period.

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-09 21:33:41	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:33:42	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:33:45	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:33:46	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:33:52	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (1m)
2026-09-09 21:33:52	10.0.0.17	DDoS	SYN Flood	92.9%	Critical	Time Ban (1m)
2026-09-09 21:33:52	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-09 21:33:53	10.0.0.17	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-09 21:33:56	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-09 21:33:56	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-09 21:33:56	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-09 21:33:57	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-09 21:34:12	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:13	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:15	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:23	10.0.0.20	DDoS	UDP Flood	76.6%	High	Time Ban (1m)
2026-09-09 21:34:23	10.0.0.20	DDoS	UDP Flood	76.6%	High	Time Ban
2026-09-09 21:34:24	10.0.0.21	DDoS	UDP Flood	79.6%	High	Time Ban (1m)
2026-09-09 21:34:24	10.0.0.21	DDoS	UDP Flood	79.6%	Critical	Time Ban
2026-09-09 21:34:26	10.0.0.22	DDoS	UDP Flood	79.6%	High	Time Ban (1m)
2026-09-09 21:34:26	10.0.0.22	DDoS	UDP Flood	79.6%	High	Time Ban
2026-09-09 21:34:42	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:43	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:50	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:52	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (2m)
2026-09-09 21:34:53	10.0.0.17	DDoS	SYN Flood	92.9%	Critical	Time Ban (2m)
2026-09-09 21:34:53	10.0.0.23	DDoS	ICMP Flood	91.9%	High	Time Ban (1m)
2026-09-09 21:34:55	10.0.0.24	DDoS	ICMP Flood	91.6%	High	Time Ban (1m)
2026-09-09 21:34:55	10.0.0.24	DDoS	ICMP Flood	91.6%	Critical	Time Ban
2026-09-09 21:34:57	10.0.0.19	DDoS	SYN Flood	92.8%	Critical	Time Ban (2m)
2026-09-09 21:34:57	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban (2m)
2026-09-09 21:34:59	10.0.0.23	DDoS	ICMP Flood	91.9%	Critical	Time Ban
2026-09-09 21:35:00	10.0.0.25	DDoS	ICMP Flood	91.7%	Critical	Time Ban (1m)
2026-09-09 21:35:02	10.0.0.25	DDoS	ICMP Flood	91.7%	Critical	Time Ban
2026-09-09 21:39:08	10.0.0.17	DDoS	SYN Flood	92.6%	Critical	Time Ban (5m)
2026-09-09 21:40:48	10.0.0.8	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-09 21:41:50	10.0.0.8	DDoS	Uncertain	68.7%	Low	Released

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-09 21:44:40	10.0.0.19	DDoS	SYN Flood	93.0%	Critical	Time Ban (5m)
2026-09-09 21:50:43	10.0.0.16	DDoS	SYN Flood	93.2%	High	Released
2026-09-09 21:50:45	10.0.0.19	DDoS	SYN Flood	94.2%	Critical	Released
2026-09-09 21:50:55	10.0.0.1	DDoS	Uncertain	59.0%	Low	Released
2026-09-09 21:50:59	10.0.0.4	DDoS	Uncertain	58.9%	Low	Released
2026-09-09 21:51:02	10.0.0.3	DDoS	Uncertain	58.9%	Low	Released
2026-09-09 21:55:30	10.0.0.18	DDoS	SYN Flood	94.2%	Critical	Released

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE